

risk of misinterpretation and ensures that development starts on a solid foundation. Claude Cowork enhances these activities by facilitating real-time collaboration, organising brainstorming sessions, and summarising discussions into structured requirement documents. Its ability to synthesise information from multiple sources enables teams to reach consensus more efficiently. GitHub Copilot, while primarily focused on coding, assists in prototyping by generating sample code snippets based on high-level requirement descriptions, enabling rapid validation of ideas during the discovery phase.

Design and documentation

Moving into the design and documentation stages, AI further demonstrates its transformative potential. Google Antigravity automatically generates architectural diagrams and design artefacts by interpreting requirement documents and user stories, ensuring alignment between business objectives and technical implementation. This capability not only accelerates the design process but also ensures traceability and consistency across the project. Claude Cowork acts as an intelligent scribe, recording design meetings, capturing key decisions, and converting these discussions into formal design documents. It also offers contextual suggestions, drawing upon industry best practices and previous project experiences. GitHub Copilot contributes by generating boilerplate code, configuration files, and even inline comments, thereby reducing the documentation burden on developers and ensuring that codebases remain well-annotated and maintainable.

Application development

The coding and implementation phase is perhaps where AI's impact is most visible. GitHub Copilot stands out by providing real-time code suggestions, auto-completing functions, and even writing entire modules based

on brief prompts. This not only accelerates development but also reduces the likelihood of errors and inconsistencies. Google Antigravity further augments development by integrating with diverse code repositories and orchestrating automated builds, ensuring that the latest code is always tested and ready for deployment. Claude Cowork supports developers by offering contextual advice, flagging potential issues, and facilitating peer reviews, all within a collaborative environment that bridges the gap between remote and in-person teams.

Test case and strategy development

Testing and validation, long considered bottlenecks in the SDLC, are being redefined by AI-driven strategies. Google Antigravity automatically generates comprehensive test cases from

requirement documents and user stories, ensuring that every business rule is thoroughly validated. It also orchestrates test execution across multiple environments, collecting and analysing results to pinpoint areas of concern. Claude Cowork supports quality assurance by facilitating the creation of test plans, managing test data, and coordinating defect triage meetings. Its collaborative features ensure that all stakeholders are kept in the loop, reducing delays and miscommunication. GitHub Copilot assists developers by suggesting unit and integration test code, thereby embedding quality directly into the development workflow and enabling test-driven development with minimal overhead.

Operations and monitoring

The operational phase of the SDLC benefits immensely from AI's predictive and analytical capabilities.

Feature	Google Antigravity	Claude Cowork	GitHub Copilot
Requirement analysis	Automated extraction from conversations and documents	Collaborative brainstorming and structured documentation	Prototype code generation from requirements
Design automation	Generates architectural diagrams and artefacts	Records design meetings and formalises documents	Boilerplate code and configuration generation
Development assistance	Automated build orchestration and code integration	Contextual advice and peer review facilitation	Real-time code suggestions and auto-completion
Testing and QA	Automated test case generation and execution	Test plan management and defect coordination	Unit and integration test code suggestions
Operations and monitoring	Deployment automation and anomaly detection	Aggregates logs, feedback, and incident analysis	Deployment scripts and monitoring dashboard generation
Collaboration	Integrates with multiple tools and platforms	Optimised for distributed team interactions	Integrates with code editors and version control